A　【基本】次の式を展開しなさい。

(1) $(x-8)(x-2) = x^2-10x+16$

(2) $(x+9)^2 = x^2+18x+81$

(3) $-2x(x+3) = -2x^2-6x$

(4) $(x+10)(x-10) = x^2-100$

(5) $(a-1)(a+8) = a^2+7a-8$

(6) $(a-8)^2 = a^2-16a+64$

(7) $(y-7)(y+5) = y^2-2y-35$

(8) $(a+4)(a-4) = a^2-16$

(9) $(m-3)(m+8) = m^2+5m-24$

(10) $(y+3)^2 = y^2+6y+9$

(11) $-2a(a+7) = -2a^2-14a$

(12) $(y+5)(y-5) = y^2-25$

(13) $(b-3)(b+9) = b^2+6b-27$

(14) $(m-4)^2 = m^2-8m+16$

(15) $3y(y+1) = 3y^2+3y$

(16) $(n-3)(n+8) = n^2+5n-24$

A　【基本】次の式を因数分解しなさい。

(1) $x^2+4x-32 = (x-4)(x+8)$

(2) $7x-35y = 7(x-5y)$

(3) $a^2-25 = (a+5)(a-5)$

(4) $x^2+18x+81 = (x+9)^2$

(5) $a^2-10a+24 = (a-6)(a-4)$

(6) $y^2-3y-18 = (y-6)(y+3)$

(7) $y^2-36 = (y+6)(y-6)$

(8) $a^2-10a+25 = (a-5)^2$

(9) $m^2-6m-7 = (m-7)(m+1)$

(10) $5ab-30b = 5b(a-6)$

(11) $m^2-81 = (m+9)(m-9)$

(12) $y^2-4y+4 = (y-2)^2$

(13) $b^2+10b+16 = (b+2)(b+8)$

(14) $n^2-2n-15 = (n-5)(n+3)$

(15) $5y-25z = 5(y-5z)$

(16) $m^2-14m+49 = (m-7)^2$

B　【基本】次の式を因数分解しなさい。

(1) $x^2+x-6 = (x-2)(x+3)$

(2) $3m-15n = 3(m-5n)$

(3) $b^2-4 = (b+2)(b-2)$

(4) $b^2-16b+64 = (b-8)^2$

(5) $a^2+8a+12 = (a+2)(a+6)$

(6) $y^2+6y+5 = (y+1)(y+5)$

(7) $n^2-4 = (n+2)(n-2)$

(8) $n^2-4n+4 = (n-2)^2$

(9) $m^2+2m-24 = (m-4)(m+6)$

(10) $b^2-2b = b(b-2)$

(11) $m^2-36 = (m+6)(m-6)$

(12) $x^2+14x+49 = (x+7)^2$

(13) $b^2-6b+8 = (b-4)(b-2)$

(14) $n^2+7n+10 = (n+2)(n+5)$

(15) $n^2-8n = n(n-8)$

(16) $a^2+6a+9 = (a+3)^2$

C　【基本】次の式を因数分解しなさい。

(1) $x^2+x-30 = (x-5)(x+6)$

(2) $2x+10y = 2(x+5y)$

(3) $b^2-100 = (b+10)(b-10)$

(4) $y^2+12y+36 = (y+6)^2$

(5) $a^2-6a-16 = (a-8)(a+2)$

(6) $y^2+5y-24 = (y-3)(y+8)$

(7) $9-x^2 = (3+x)(3-x)$

(8) $m^2+14m+49 = (m+7)^2$

(9) $m^2+3m-18 = (m-3)(m+6)$

(10) $8a+24b = 8(a+3b)$

(11) $49-a^2 = (7+a)(7-a)$

(12) $b^2+14b+49 = (b+7)^2$

(13) $b^2+4b-5 = (b-1)(b+5)$

(14) $n^2+10n+9 = (n+1)(n+9)$

(15) $3y^2+24y = 3y(y+8)$

(16) $n^2+18n+81 = (n+9)^2$